Carnegie Mellon University
Master of
Software Engineering

Architectural drivers

LG Architecture Training Program

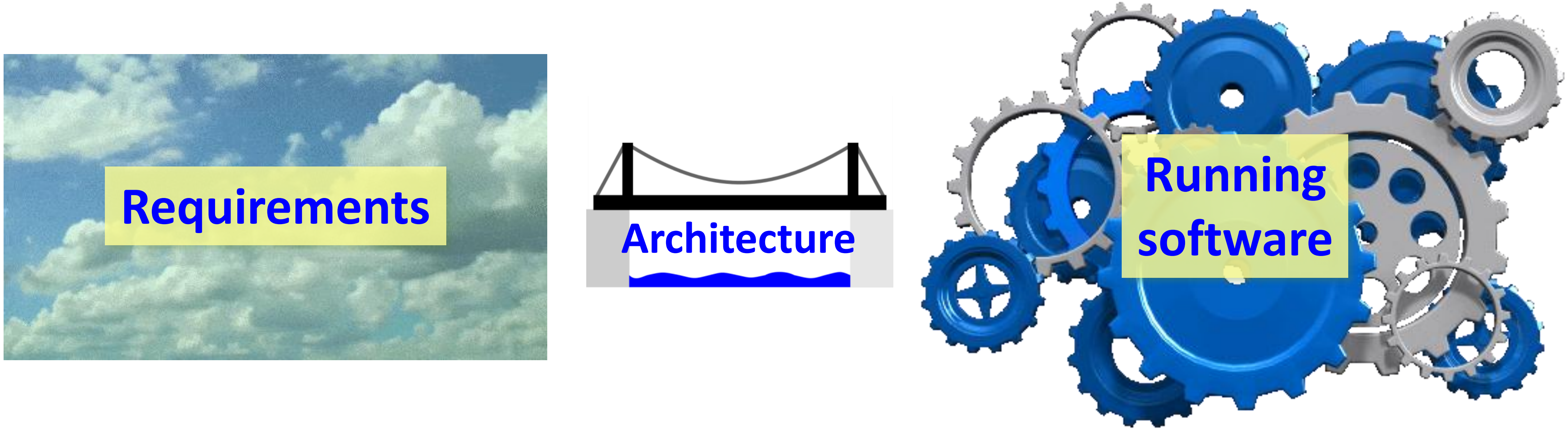
Paulo Merson

Motivation for Software Architecture (1)

The challenge of creating software

Motivation for Software Architecture (2)

The role of software architecture



Types of requirements

- There are 3 types of requirements
  - **Functional requirements**: what the system shall do, how it should react to stimulus
  - **Quality attribute requirements**: performance, usability, modifiability, etc.
  - **Design constraints**: a design decision already made; it can't be changed
- Functional requirements can be expressed, for example, as
  - Use cases (UML use case diagram + use case description)
  - User stories

Quality attribute requirements are aka non-functional requirements (NFR)

Architectural drivers

- A requirement of any type can influence the architecture
- Requirements that influence the architecture are called
  - *architectural drivers* or
  - *architecturally significant requirements (ASRs)*
- QA requirements are essential input to define
  - architecture structures
  - design patterns to be used
  - important design decisions

In this lecture and most places in this course, QA means quality attribute, not quality assurance 🙃

QA requirements and design decisions

Quality attribute requirements have great influence in the architecture

- Examples (for a distributed system):

Quality requirement	Design decision
Scalability	Deploy the *Version Update* service as an AWS lambda function
Security	Use OIDC authentication in the *Checkout* service
Performance	Use data replication with a MongoDB database for the *Report* service
Modifiability	Structure the code in layers: presentation, service, domain
Deployability	Use Docker to deploy microservices in the *Auth* system
Interoperability	Migrate SOAP services to REST

We need to specify these rqmts in a more precise way

A common mistake: focus on functional rqmts

If functionality were the only thing that mattered,
you wouldn't have to divide the system into pieces at all;
a single monolithic blob with no internal structure would do just fine. [Bass21]

Nonetheless, people often pay too much attention to functionality and overlook QA requirements

Len Bass

Questions?

Paulo Merson
pmerson@cmu.edu